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Lab Procedure
Time of Flight

Introduction

Ensure the following:

1. You have reviewed the Application Guide – Time of Flight
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
    a. The LED next to the USB C port will stay white after the touch screen starts loading.
    b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

Exploring Time of Flight Sensors

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File** > **Open Folder**..., and browse to the directory containing the python files and lab procedure for the Time-of-Flight lab (.../sensors_trainer/1_fundamentals /time_of_flight/hardware/python). Open this directory.
2. Open **TOF_single_beam.py**, this file will be used for the first experiment.
3. Although the sensor gives an 8x8 grid of depth data, we will just be focusing on one of those points for this section. To understand how data is represented open the provided OPT-VL53L8CHV0GC-1.pdf located in (.../sensors_trainer/ 1_fundamentals/time_of_flight). In page 26 we get a view of how the raw data is indexed from 0-15 (for a 4x4 grid) and flipped. We flip and reconstruct the measurement grid to get a better understanding of which single point measurement we are studying for this portion of the lab.
4. In this experiment we'll explore three important measurements provided by the Time of Flight Sensor (VL53L8CHV0GC). They are:
    a. Distance to object (m)
    b. Distance Sigma (m)
    c. Reflectance of object (%)
5. Place a cardboard surface (or a matte flat surface such as the back of a notebook, a binder or a laptop case) 0.5m away from the Mechatronics Sensors Trainer board.

6. Click ▷ to run the file.
7. Slowly move the cardboard surface closer to the device, up to 0.1m away from it. Take screenshots of your plots.
8. What do you notice about the distance, sigma and reflectance plots?
9. Run steps 5-8 and this time redo the experiment with a reflective surface (You could use the front screen on a smartphone) type material.
10. How do the distance, sigma, and reflectance plots change now?
11. This time repeat steps 5-10 except as you bring the surface closer to the ToF sensor oscillate the surface back and forth.
12. Do the distance, sigma, and reflectance values vary in the same way as the first time the experiment was ran?
13. The second experiment will look at TOF_3D.py. In This example we will consider a 3D view of an object in space. Note that for this example to run, due to the use of open3d pip package, the python version you should be using is 3.12 or older. Based on our experience we had try to install open3d around 3 times for it to install successfully. Try using the **`pip install open3d`** command for this.
14. Place a large flat surface 0.3m away from the Mechatronics Sensor Trainer. Place a roll of tape (or a donut shaped object) 0.15m away from the Mechatronics Sensor Trainer. Make sure the tape is placed on the round edge and not the flat edge as so to cast a circular shadow on the flat surface.
15. Click ▷ to run the experiment.
16. Redo steps 14-15 with a clear object (ex: clear water bottle). Does this affect the distance measurement?
17. What do you notice about the 3D point cloud? What type of geometric objects can a multizone Time of Flight sensor identify if you're visualizing the 3D point cloud?
18. For the third experiment, keep the experimental setup the same and run TOF_colorized.py. This file will take the 8x8 distance measurements from the sensor and display them as an image.
19. What do you notice about the depth measurement for the colorized multizone measurement? Is there a limitation to this approach? What happens when the flat surface is placed 1m away from the sensor (and the roll of tape is place 0.5m away)?
20. Stop, save and close your program.
21. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.